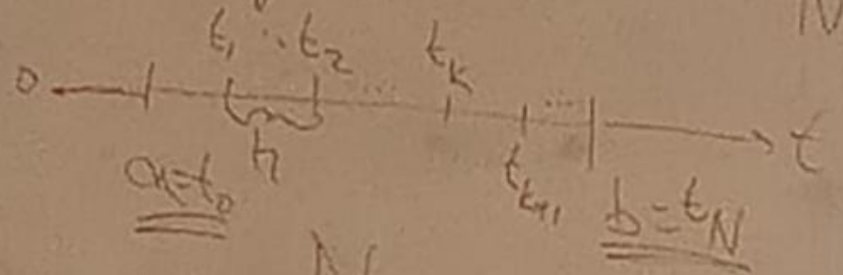
Euler Yöntemi

Diferansiyel Denklemler için Sayısal Yöntemler

- Euler Yöntemi
- Yüksek dereceden Taylor Yöntemi

var + tek

$$\begin{cases} y' = f(t,y), \\ a \le t \le b \\ y(a) = \alpha \end{cases}$$



N parçaya bölü

Adım Ölçüsü. $h = \frac{b-a}{N} = t_{k+1} - t_k$

$t_k = a + kh \quad k = 0, 1, \ldots, N$

$$\begin{cases} y' = 1 + \frac{y}{t}, & 1 \le t \le 2 \\ y(1) = 2 \end{cases}$$

Bu başlangıç değer prob. çözümünün tahmini değerini adım boyu $h = 0.25$ olacak biçimde belirleyiniz.

EULER YÖNTEMİ

$w_0 = \alpha = 2$

$w_{k+1} = w_k + 0.25\left(1 + \frac{w_k}{t_k}\right)$

$w_{k+1} = w_k + h\,f(t_k, w_k)$

$k = 0, 1, \ldots, N-1$

$= 0, 1, 2, 3$

$f(t,y) = 1 + \frac{y}{t}$, $a = 1, b = 2$, $\alpha = 2$, $h = \frac{b-a}{N} \Rightarrow N = \frac{b-a}{h} = 4$

$k=0,\ w_1 = w_0 + 0.25\left(1 + \frac{w_0}{t_0}\right)$

$k=1,\ w_2 = w_1 + 0.25\left(1 + \frac{w_1}{t_1}\right)$

$k=2,\ w_3 = w_2 + 0.25\left(1 + \frac{w_2}{t_2}\right)$

$k=3,\ w_4 = w_3 + 0.25\left(1 + \frac{w_3}{t_3}\right)$

$y' - \frac{1}{t}y = 1 \Rightarrow y(t) = 2x + x\ln|x|$, $y(1) = 2$

TAHMİNİ HATA

$y(t_{k+1}) = y(t_k) + h\,y'(t_k) + \frac{h^2}{2}y''(\xi_k)$

$y'(t_k) = f(t_k, y(t_k))$

$y(t_0) = \alpha$

$y(t_{k+1}) = y(t_k) + h\,f(t_k, y(t_k))$ ← TAHMİNİ

$+ \frac{h^2}{2}f'(\xi_k)$

$k = 0, 1, \ldots, N-1$

$w_k \approx y(t_k)$

$$\begin{cases} y' = 1 + \frac{y}{t}, & 1 \le t \le 2 \\ y(1) = 2 \end{cases}$$

başlangıç değer-a prob. çözümünün tahmini değerini adım boyu $h = 0.25$ tablo biçiminde belirleyiniz.

$f(t,y) = 1 + \frac{y}{t}$, $a=1, b=2$, $\alpha=2$, $h = \frac{b-a}{N} \Rightarrow N = \frac{b-a}{h} = 4$

$y' - \frac{1}{t}y = 1$

$t_0 = 1$, $w_1 = w_0 + 0.25\left(1 + \frac{w_0}{t_0}\right)$

t_1, $w_2 = w_1 + 0.25\left(1 + \frac{w_1}{t_1}\right)$

t_2, $w_3 = w_2 + 0.25\left(1 + \frac{w_2}{t_2}\right)$

t_3, $3 + 0.25\left(1 + \frac{w_3}{t_3}\right)$

EULER YÖNTEMİ

$w_0 = \alpha = 2$

$w_{k+1} = w_k + 0.25\left(1 + \frac{w_k}{t_k}\right)$

$w_{k+1} = w_k + h f(t_k, w_k)$

$k = 0, 1, \ldots, N-1$

$= 0, 1, 2, 3$

TAHMİNİ HATA

$y(t_{k+1}) = y(t_k) + h y'(t_k) + \frac{h^2}{2} y''(\xi_k)$

$y'(t_k) = f(t_k, y(t_k))$

$y(t_0) = \alpha$

$y(t_{k+1}) = y(t_k) + h f(t_k, y(t_k)) + \frac{h^2}{2} f'(\xi_k)$

$k = 0, 1, \ldots, N-1$

$w_i \approx y(t_k)$

t_k; $k = 0, 1, \ldots, N-1$

$y \in C^{(2)}[a,b]$ · $t < \xi_k$

$y(t) = y(t_k) + y'(t_k)(t - t_k) + \frac{y''(\xi_k)}{2!}(t - t_k)^2$

$t = t_{k+1}$

$y(t_{k+1}) = y(t_k) + y'(t_k)(t_{k+1} - t_k) + \frac{y''(\xi_k)}{2}(t_{k+1} - t_k)^2$

$$\begin{cases} y' = 1 + \frac{y}{t}, & 1 \le t \le 2 \\ y(1) = 2 \end{cases}$$

Bu başlangıç değer prob. çözümünün tahmini değerini adım boyu $h = 0.25$ olacak biçimde belirleyiniz

EULER YÖNTEMİ

$w_0 = \alpha = 2$

$w_{k+1} = w_k + 0.25\left(1 + \frac{w_k}{t_k}\right)$

$w_{k+1} = w_k + h\, f(t_k, w_k)$

$k = 0, 1, \ldots, N-1$

$= 0, 1, 2, 3$

$f(t,y) = 1 + \frac{y}{t}$, $a=1, b=2, \alpha=2, h = \frac{b-a}{N} \Rightarrow N = \frac{b-a}{h} = 4$

$k=0,\ w_1 = w_0 + 0.25\left(1 + \frac{w_0}{t_0}\right)$

$k=1,\ w_2 = w_1 + 0.25\left(1 + \frac{w_1}{t_1}\right)$

$k=2,\ w_3 = w_2 + 0.25\left(1 + \frac{w_2}{t_2}\right)$

$k=3,\ w_4 = w_3 + 0.25\left(1 + \frac{w_3}{t_3}\right)$

TAYLOR YÖNTEMİ

$y(t_{k+1}) = y(t_k) + h\,y'(t_k) + \frac{h^2}{2} y''(\xi_k)$

$y'(t_k) = f(t_k, y(t_k))$

$y(t_0) = \alpha$

$y(t_{k+1}) = y(t_k) + h f(t_k, y(t_k)) + \frac{h^2}{2} f'(\xi_k)$

$k = 0, 1, \ldots, N-1$

$w_k \approx y(t_k)$

t_k, $k = 0, 1, \ldots, N-1$

$y \in C^{(2)}[a,b]$ · $t < \xi_k < t_k$

$y(t) = y(t_k) + y'(t_k)(t - t_k) + \frac{y''(\xi_k)}{2!}(t - t_k)^2$

$t = t_{k+1}$

$y(t_{k+1}) = y(t_k) + y'(t_k)\frac{(t_{k+1} - t_k)}{h} + \frac{y''(\xi_k)}{2}\frac{(t_{k+1} - t_k)^2}{h^2}$

$$\begin{cases} y' = 1 + \frac{y}{t} & , \quad 1 \le t \le 2 \\ y(1) = 2 \end{cases}$$

Bu başlangıç değer problemini çözümünün tahmini değerini, adım boyu $h = 0.25$ olacak biçimde belirleyiniz.

$$f(t,y) = 1 + \frac{y}{t} \quad ; \quad a=1, b=2, \quad \alpha = 2, \quad h = \frac{b-a}{N} \Rightarrow N = \frac{b-a}{h} = 4$$

EULER YÖNTEMİ

$w_0 = \alpha$

$w_{k+1} = w_k + h f(t_k, w_k)$

$k = 0, 1, \ldots, N-1$

TAYLOR YÖNTEMİ

$$y(t_{k+1}) = y(t_k) + h y'(t_k) + \frac{h^2}{2} y''(\xi_k)$$

$$y'(t_k) = f(t_k, y(t_k))$$

$$y(t_0) = \alpha$$

$$y(t_{k+1}) = y(t_k) + h f(t_k, y(t_k)) + \frac{h^2}{2} f'(\xi_k)$$

$k = 0, 1, \ldots, N-1$

$w_k \approx y(t_k)$

$t_k = \quad k = 0, 1, \ldots, N$

$y \in C^{(2)}[a,b]$ -

$y(t) = y(t_k) + y$

$+ \frac{y''(\xi_k)}{2!}$

$t = t_{k+1}$

$y(t_{k+1}) = y(t_k) +$

$+ y''(\xi$

$$\begin{cases} y' = 1 + \frac{y}{t}, & 1 \le t \le 2 \\ y(1) = 2 \end{cases}$$

Bu başlangıç değer problemin
tahmini değerinin adım
olacak biçimde belirten

$f(t,y) = 1 + \frac{y}{t}$, $\alpha = 2$, $h = \frac{b-a}{N} \Rightarrow N = \frac{b-a}{h} = 4$

EULER YÖNTEMİ

$w_0 = \alpha$

$w_{k+1} = w_k + h f(t_k, w_k)$

$k = 0, 1, \dots, N-1$

TAHMİN HATA

$y(t_{k+1}) = y(t_k) + h y'(t_k) + \frac{h^2}{2} y''(\xi_k)$

$y'(t_k) = f(t_k, y(t_k))$

$y(t_0) = \alpha$

$y(t_{k+1}) = y(t_k) + h f(t_k, y(t_k)) + \frac{h^2}{2} f'(\xi_k)$

TAHMİN

$k = 0, 1, \dots, N-1$

$w_k \approx y(t_k)$

t_k , $k = 0, 1, \dots, N-$

$y \in C^{(2)}[a,b]$ - t^2

$y(t) = y(t_k) + y'(t$

$t = t_{k+1}$ $+ \frac{y''(\xi_k)}{2!}(t$

$y(t_{k+1}) = y(t_k) + y'(t$

$+ y''(\xi_k$

2

$$\begin{cases} y' = 1 + \frac{y}{t}, & 1 \le t \le 2 \\ y(1) = 2 \end{cases}$$

EULER YÖNTEMİ

$w_0 = \alpha$

$w_{k+1} = w_k + h f(t_k, w_k)$

$k = 0, 1, \dots, N-1$

TAHMİN HATA

$y(t_{k+1}) = y(t_k) + h y'(t_k) + \frac{h^2}{2} y''(\xi_k)$

$y'(t_k) = f(t_k, y(t_k))$

$y(t_0) = \alpha$

$y(t_{k+1}) = y(t_k) + h f(t_k, y(t_k)) + \frac{h^2}{2} f'(\xi_k)$

$k = 0, 1, \dots, N-1$

$w_k \approx y(t_k)$

$t_k = , k = 0, 1, \dots$

$y \in C^{(2)}[a, b]$ -

$y(t) = y(t_k) + y$

$t = t_{k+1}$

$+ \frac{y''(\xi_k)}{2!}$

$y(t_{k+1}) = y(t_k) + $

$+ y''($

Euler Yöntemi

Diferansiyel Denklemler için Sayısal Yöntemler

- Euler Yöntemi
- Yüksek dereceden Taylor Yöntemi

t_k ; $k=0,1,\dots,N-1$

$y \in C^{(2)}[a,b]$ — $t < \xi_k < t_k$

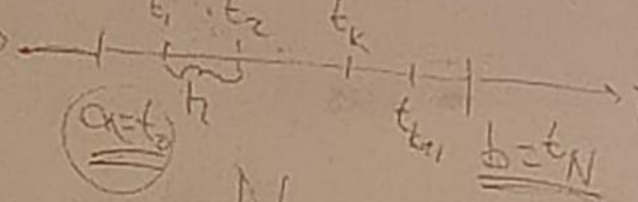
$$y(t) = y(t_k) + y'(t_k)(t-t_k) + \frac{y''(\xi_k)}{2!}(t-t_k)^2$$

$t = t_{k+1}$

$$y(t_{k+1}) = y(t_k) + y'(t_k)\underbrace{(t_{k+1}-t_k)}_{h} + \frac{y''(\xi_k)}{2}\underbrace{(t_{k+1}-t_k)^2}_{h^2}$$

var + tek

$$\begin{cases} y' = f(t,y), \\ a \le t \le b \\ y(a) = \alpha \end{cases}$$



N parçaya böl

Adım Ölçüsü: $h = \frac{b-a}{N} = t_{k+1} - t_k$

$t_k = a + kh \quad k = 0,1,\dots,N$

$$y(t_{k+1}) = y(t_k) + h\,y'(t_k) + \frac{h^2}{2}y''(\xi_k)$$

Yüksek dereceden Taylor Yöntemi

$$\frac{y(t_{k+1}) - y(t_k)}{h} - y'(t_k) = \frac{\frac{h}{2}y''(\xi_k)}{}$$

$$t_k < \xi_k < t_{k+1}$$

$y \in C^2[a,b]$

$$\left|\frac{y''(\xi_k)}{2}\right| \le M$$

$$|\tau_{k+1}(h)| \le Mh$$

$O(h)$

Diferansiyel Denklemler için Sayısal Yöntemler

- Euler Yöntemi
- Yüksek dereceden Taylor Yöntemi

$$\begin{cases} y' = f(t,y), & a \le t \le b \\ y(a) = \alpha \end{cases}$$

Euler yöntemi $y(t_k) = w_k$

$w_0 = \alpha$

$$w_{k+1} = w_k + h f(t_k, w_k), \quad k = 0, 1, \dots, N-1$$

Sürecin lokal kesim hatası, $y(t_k) = y_k$ $k = 0, 1, \dots, N$

$h = t_{k+1} - t_k$

$N = \frac{b-a}{h}$ Adım ilerisi

$$\tau_{k+1}(h) = \frac{y_{k+1} - y_k}{h} - y'(t_k) = \frac{y''(\xi_k)}{2}h$$

$a = t_0$ t_1 t_2 $\dots$ t_k t_{k+1} $\dots$ t_{N-1} $b = t_N$

$w_0 = \alpha$

$w_{k+1} = w_k + h T^{(n)}(t_k, w_k)$, $k = 0, 1, \dots, N-1$

$LKH \rightarrow |\tau_{k+1}(h)| \leq M_1 h^n$

$O(h^n)$

$n = 2$ συνηθισμένη

$T^{(2)}(t_k, w_k) = f(t_k, w_k) + \frac{h}{2} \frac{\partial}{\partial t} f(t_k, w_k)$

$y(t_{k+1}) = y(t_k) + h y'(t_k) + \frac{h^2}{2} y'(t_k) + \dots + \frac{h^n}{n!} y^{(n)}(t_k)$

$+ \frac{h^{n+1}}{(n+1)!} y^{(n+1)}(\xi_k)$, $t_k \leq \xi_k \leq t_{k+1}$

$y'(t_k) = f(t_k, y_k)$, $y''(t_k) = \frac{d}{dt} f(t_k, y(t_k))$

$y^{(n)}(t_k) = \frac{d}{dt^n} f(t_k, y(t_k))$

w_{k+1} w_k

$y(t_{k+1}) = y(t_k) + h f(t_k, y(t_k)) + \frac{h^2}{2} \frac{d}{dt} f(t_k, y(t_k)) + \dots$

$\dots + \frac{h^n}{n!} \frac{d^{n-1}}{dt^{n-1}} f(t_k, y(t_k)) + \frac{h^{n+1}}{(n+1)!} \frac{d^n}{dt^n} f(\xi_k, y(\xi_k))$

$T^{(n)}(t_k, w_k)$

ΑΤΑΗ

$\begin{cases} y' = y - t^2 + 1, & 0 \le t \le 1 \\ y(0) = 0.5 \end{cases}$ $f(t,y) = y - t^2 + 1$ $a=0, b=1, \alpha=0.5$

$\frac{d}{dt} f(t,y(t)) = y' - 2t = y - t^2 - 2t + 1$

Yüksek dereceden Taylor Yöntemi

BDP 2. den Taylor formülünü adım boyu $h=0.25$ olarak yaklaşık çözün.

$N = \frac{b-a}{h} = 4$ $h = \frac{1}{4}$

$t_k = a + kh = 0.25\,k = \frac{k}{4}$ $k = 0,1,2,3,4$

$y' = f(t,y), \quad a \le t \le b$

$y(a) = \alpha$

$n=2$ için Taylor formülü

$w_0 = 0.5$

$w_{k+1} = w_k + 0.25\, T^{(2)}(t_k, w_k)$

Diferansiyel Denklemler için Sayısal Yöntemler

- Euler Yöntemi
- Yüksek dereceden Taylor Yöntemi

$w_0 = \alpha$

$w_{k+1} = w_k + h T^{(n)}(t_k, w_k)$, $k = 0, 1, \dots, N-1$

LKH $\rightarrow |\tau_{k+1}(h)| \le M h^n$

$O(h^n)$

$n = 2$ durumunda

$T^{(2)}(t_k, w_k) = f(t_k, w_k) + \frac{h}{2} f'(t_k, w_k)$

$= w_k - t_k^2 + 1 + \frac{h}{2}\left(w_k - t_k^2 - 2t_k + 1\right)$

$= w_k\left(1 + \frac{1}{4}\right) + t_k^2 ($

$y(t_{k+1}) = y(t_k) + h y'(t_k) + \frac{h^2}{2} y''(t_k) + \dots + \frac{h^n}{n!} y^{(n)}(t_k) + \frac{h^{n+1}}{(n+1)!} y^{(n+1)}(\xi_k)$, $t_k \le \xi_k \le t_{k+1}$

$y'(t_k) = f(t_k, y_k)$, $y''(t_k) = \frac{d}{dt} f(t_k, y(t_k))$

$y^{(n)}(t_k) = \frac{d}{dt^n} f(t_k, y(t_k))$

$y(t_{k+1}) = y(t_k) + h f(t_k, y(t_k)) + \frac{h^2}{2} \frac{d}{dt} f(t_k, y(t_k)) + \dots + \frac{h^n}{n!} \frac{d^{n-1}}{dt^{n-1}} f(t_k, y(t_k)) + \frac{h^{n+1}}{(n+1)!} \frac{d^n}{dt^n} f(\xi_k, y(\xi_k))$

w_{k+1} w_k

HATA

$T^{(n)}(t_k, w_k)$

$\begin{cases} y' = y - t^2 \\ y(0) = \end{cases}$

BDP 2

olarak

$t_k = a +$

$y' = y - t^2 + 1, \quad 0 \le t \le 2$

$y(0) = 0.5$

$f(t,y) = y - t^2 + 1$

$a = 0, b = 1, \alpha = 0.5$

$\frac{d}{dt} f(t,y(t)) = y' - 2t = y - t^2 - 2t + 1$

Yüksek dereceden Taylor Yöntemi

BDP ? dan Taylor formülüne adım boyu $h = 0.25$ olarak yaklaşık çözüm!

$N = \frac{b-a}{h} = 4$

$t_k = a + kh = 0.25\,k = \frac{k}{4}$

$k = 0, 1, 2, 3, 4$

$k = 0 \Rightarrow w_1 = 0.9219$

$k = 1 \Rightarrow w_2 = 1.4292$

$k = 2 \Rightarrow w_3 = 2.0169$

$k = 3 \Rightarrow w_4 = 2.6526$

$y' = f(t,y), \quad a \le t \le b$

$y(a) = \alpha$

$n = \frac{1}{4}$

n=2 için Taylor formülü

$w_0 = 0.5$

$w_{k+1} = w_k + 0.25\, T^{(2)}(t_k, w_k)$

$w_{k+1} = w_k + \frac{1}{4}\left(\frac{9}{8}\left(w_k - \frac{1}{16}\right) + 1\right)$

Diferansiyel Denklemler için Sayısal Yöntemler

- Euler Yöntemi
- Yüksek

$y' = y - t^2 + 1$

$y(t_k) = y(t_{k-1}) + h f(t_{k-1}, y(t_{k-1})) + \frac{h^2}{2} f'(t_{k-1}, y(t_{k-1})) + \cdots + \frac{h^n}{n!} f^{(n)}(t_{k-1}) + \frac{h^{n+1}}{(n+1)!} y^{(n+1)}(\xi_k), \quad t_{k-1} \le \xi_k \le t_{k+1}$

$y'(t_k) = f(t_k, y(t_k))$, $y''(t_k) = \frac{d}{dt} f(t_k, y(t_k))$

$y^{(n)}(t_k) = \frac{d^{n-1}}{dt^{n-1}} f(t_k, y(t_k))$

$f(t_k, y(t_k)) + \frac{h^2}{2} \frac{d}{dt} f(t_k, y(t_k)) + \cdots + \frac{h^{n+1}}{(n+1)!} \frac{d^n}{dt^n} f(\xi_k, y(\xi_k))$

HATA

(t_k, w_k)

$\begin{cases} y' = y - t^2 + 1, & 0 \le t \le 1 \\ y(0) = 0.5 \end{cases}$ $\begin{cases} f(t,y) = y - t^2 + 1 \\ a=0, b=1, \alpha = 0.5 \end{cases}$ Yüksek dereceden Taylor Yöntemi

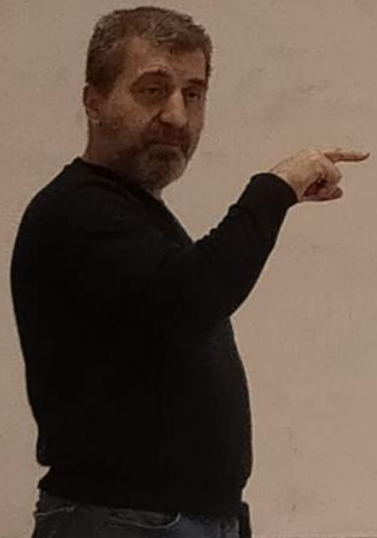
$\frac{df}{dt}(t,y(t)) = y' - 2t = y - t^2 - 2t + 1$

BDP ? deri Taylor formülünü adım boyutu h olarak yaklaşık çözün!

$\begin{cases} y' = f(t,y), & a \le t \le b \\ \end{cases}$

Diferansiyel Denklemler için Sayısal Yöntemler

- Euler Yöntemi
- Yüksek dereceden Taylor Yöntemi



$$y(t_{k+1}) = y(t_k) + h\,y'(t_k) + \frac{h^2}{2}y''(t_k) + \cdots + \frac{h^n}{n!}y^{(n)}(t_k) + \frac{h^{n+1}}{(n+1)!}y^{(n+1)}(\xi_k), \quad t_k \le \xi_k \le t_{k+1}$$

$$y'(t_k) = f(t_k, y_k), \quad y''(t_k) = \frac{d}{dt}f(t_k, y(t_k))$$

$$y^{(n)}(t_k) = \frac{d^n}{dt^n}f(t_k, y(t_k))$$

$$y(t_{k+1}) = y(t_k) + h f(t_k, y(t_k)) + \frac{h^2}{2}\frac{d}{dt}f(t_k, y(t_k)) + \cdots + \frac{h^n}{n!}\frac{d^{n-1}}{dt^{n-1}}f(t_k, y(t_k)) + \underbrace{\frac{h^{n+1}}{(n+1)!}\frac{d^n}{dt^n}f(\xi_k, y(\xi_k))}_{\text{HATA}}$$

$$y(t_{k+1}) = y(t_k) + h\,y'(t_k) + \frac{h^2}{2}y''(\xi_k)$$

$$\frac{y(t_{k+1}) - y(t_k)}{h} - y'(t_k) = \frac{\frac{h}{2}y''(\xi_k)}{}, \quad t_k < \xi_k < t_{k+1}$$

$y \in C^2[a,b]$

$$\left|\frac{y''(\xi_k)}{2}\right| \le M$$

$$|\tau_{k+1}(h)| \le M\,h$$

$O(h)$

Yüksek dereceden

$$\begin{cases} y' = f(t \\ y(a) = \end{cases}$$

Euler yaklaşımı

$w_0 = \alpha$

$w_{k+1} = w_k +$

Sürecin

$h = t_{k+1} - t_k$

$N = \frac{b-a}{h}$

Ardışıkları

$\tau_k(h)$

$a = t_0 \quad t_1 \quad t_2 \quad t_k \quad t_{k+1} \quad t_{N-1} \quad b = t_N$

$y(t_{k+1}) = y(t_k) + h\,y'(t_k) + \frac{h^2}{2}y''(\xi_k)$ | Yüksek dereceden Taylor Yöntemi

Diferansiyel Denklemler için Sayısal Yöntemler

- Euler Yöntemi
- Yüksek dereceden Taylor Yöntemi

$$\frac{y(t_{k+1}) - y(t_k)}{h} - y'(t_k) = \frac{\frac{h}{2}y''(\xi_k)}{}$$

$t_k < \xi_k < t_{k+1}$

$y \in C^2[a,b]$

$$\left|\frac{y''(\xi_k)}{2}\right| \le M$$

$$|\tau_{k+1}(h)| \le M\,h$$

$O(h)$

$$\begin{cases} y' = f(t,y), & a \le t \le b \\ y(a) = \alpha \end{cases}$$

Euler yöntemi $y(t_k) = w_k$

$w_0 = \alpha$

$w_{k+1} = w_k + h f(t_k, w_k), \quad k = 0, 1, \dots, N-1$

Sürecin lokal kesim hatası, $(y(t_k) = y_k)$ $k = 0, 1, \dots, N$

$$\tau_{k+1}(h) = \frac{y_{k+1} - y_k}{h} - y'(t_k) = \frac{y''(\xi_k)}{2}h$$

$$w_0 = \alpha$$

$$w_{k+1} = w_k + h T^{(n)}(t_k, w_k) \quad , k = 0, 1, \dots, N-1$$

$$LKH \rightarrow |\tau_{k+1}(h)| \leq M_1 h^n$$

$O(h^n)$

$n = 2$ durumunda

$$T^{(2)}(t_k, w_k) = f(t_k, w_k) + \frac{h}{2} \frac{\partial}{\partial t} f(t_k, w_k)$$

$$y(t_{k+1}) = y(t_k) + h y'(t_k) + \frac{h^2}{2} y''(t_k) + \cdots + \frac{h^{n+1}}{(n+1)!} y^{(n+1)}(\xi_k) \quad ,$$

$$y'(t_k) = f(t_k, y_k) \quad , \quad y''(t_k) = \frac{d}{dt} f$$

$$y^{(n)}(t_k) = \frac{d^n}{dt^n}$$

$$y(t_{k+1}) = y(t_k) + h f(t_k, y(t_k)) + \frac{h^2}{2} \frac{d}{dt} f + \cdots + \frac{h^n}{n!} \frac{d^{n-1}}{dt^{n-1}} f(t_k, y(t_k)) + \frac{h^{n+1}}{(n+1)!}$$

(w_{k+1}, w_k)

$w_0 = \alpha$

$w_{k+1} = w_k + h T^{(n)}(t_k, w_k)$, $k = 0, 1, \dots, N-1$

LKH $\rightarrow |\tau_{k+1}(h)| \le M h^n$

$O(h^n)$

$n = 2$ durumunda,

$$T^{(2)}(t_k, w_k) = f(t_k, w_k) + \frac{h}{2} f'(t_k, w_k)$$

$$= w_k - t_k^2 + 1 + \frac{h}{2}\left(w_k - t_k^2 - 2t_k + 1\right)$$

$$= w_k\left(1 + \frac{1}{8}\right) + t_k^2\left(1 + \frac{1}{8}\right) - t_k\left(\frac{1}{4}\right) + \frac{9}{8} - T^{(n)}(t_k, w_k)$$

$$- \frac{9}{8} w_k - \frac{9k^2}{16} - \frac{k}{16} + \frac{9}{8}$$

$$y(t_{k+1}) = y(t_k) + h y'(t_k) + \frac{h^2}{2} y'(t_k) + \dots + \frac{h^n}{n!} y^{(n)}(t_k) + \frac{h^{n+1}}{(n+1)!} y^{(n+1)}(\xi_k) \,, \quad t_k \le \xi_k \le t_{k+1}$$

$$y'(t_k) = f(t_k, y_k) \,, \quad y''(t_k) = \frac{d}{dt} f(t_k, y(t_k))$$

$$y^{(n)}(t_k) = \frac{d^{n-1}}{dt^{n-1}} f(t_k, y(t_k))$$

w_{k+1} w_k

$$y(t_{k+1}) = y(t_k) + h f(t_k, y(t_k)) + \frac{h^2}{2} \frac{d}{dt} f(t_k, y(t_k)) + \dots + \frac{h^n}{n!} \frac{d^{n-1}}{dt^{n-1}} f(t_k, y(t_k)) + \frac{h^{n+1}}{(n+1)!} \frac{d^n}{dt^n} f(\xi_k$$

HATA

$\begin{cases} y' = y - t^2 + 1, & 0 \le t \le 1 \\ y(0) = 0.5 \end{cases}$ $\left\{ \begin{array}{l} f(t,y) = y - t^2 + 1 \\ a=0, t=1, x=0.5 \end{array} \right\}$ Yüksek dereceden Taylor Yöntemi

$\left\{ \frac{d}{dt}[f(t,y(t))] : y' - 2t = y - t^2 - 2t + 1 \right.$

BDP ? den Taylor fonksiyonu adım boyutu $h=0.2$ olarak yaklaşık çözün!

$\begin{cases} y' = f(t,y), & a \le t \le b \\ y(a) = \alpha \end{cases}$

Diferansiyel Denklemler için Sayısal Yöntemler

Yöntemi

Dereceden

Yöntemi

$y(t_{k+1}) = y(t_k) + h\,y'(t_k) + \frac{h^2}{2}y''(\xi_k)$ | Yüksek dereceden Taylor Yöntemi

Diferansiyel Denklemler için Sayısal Yöntemler

- Euler Yöntemi
- Yüksek dereceden Taylor Yöntemi

$$\frac{y(t_{k+1}) - y(t_k)}{h} - y'(t_k) = \frac{\frac{h}{2}y''(\xi_k)}{}$$

$t_k < \xi_k < t_{k+1}$

$y \in C^2[a,b]$

$$\left|\frac{y''(\xi_k)}{2}\right| \le M$$

$$|\tau_{k+1}(h)| \le Mh$$

$O(h)$

$$\begin{cases} y' = f(t,y), & a \le t \le b \\ y(a) = \alpha \end{cases}$$

Euler yöntemi $y(t_k) = w_k$

$w_0 = \alpha$

$w_{k+1} = w_k + h f(t_k, w_k)$, $k = 0, 1, \dots, N-1$

Sürecin lokal kesme hatası; $y_j(t_k) = y_k$ $k = 0, 1, \dots, N$

$$\tau_{k+1}(h) = \frac{y_{k+1} - y_k}{h} - y'(t_k) = \frac{y''(\xi_k)}{2}h$$

$$y(t_k) + \frac{h^2}{2} y''(t_k) + \cdots + \frac{h^n}{n!} y^{(n)}(t_k)$$

$$\frac{h^{n+1}}{(n+1)!} y^{(n+1)}(\xi_k), \quad t_k \le \xi_k \le t_{k+1}$$

$$y''(t_k) = \frac{d}{dt} f(t_k, y(t_k))$$

$$y^{(n)}(t_k) = \frac{d^{n-1}}{dt^{n-1}} f(t_k, y(t_k))$$

$$y(t_k)) + \frac{h^2}{2} \frac{d}{dt} f(t_k, y(t_k)) + \cdots$$

$$, y(t_k)) + \underbrace{\frac{h^{n+1}}{(n+1)!} \frac{d^n}{dt^n} f(\xi_k, y(\xi_k))}_{\text{HATA}}$$

$$\begin{cases} y' = y - t^2 + 1, & 0 \le t \le 1 \\ y(0) = 0.5 \end{cases}$$

$f(t,y) = y - t^2 + 1$

$a=0, b=1, \alpha=0.5$

$\frac{df}{dt}(t,y(t)) = y' - 2t = y - t^2 - 2t + 1$

Yüksek dereceden Taylor Yöntemi

BDP 2. den Taylor formülünü adım boyu $h=0.25$ olarak yaklaşık çözün!

$N = \frac{b-a}{h} = 4$ $\quad h = \frac{1}{4}$

$t_k = a + kh = 0.25\,k = \frac{k}{4}$ $\quad k = 0,1,2,3,4$

$k=0 \Rightarrow w_1$

$k=1 \Rightarrow w_2$

$k=2 \Rightarrow w_3$

$k=3 \Rightarrow w_4$

$$\begin{cases} y' = f(t,y), & a \le t \le b \\ y(a) = \alpha \end{cases}$$

$n=2$ ... formülü

$w_0 = 0.$

$w_{k+1} = \left(\frac{k^2}{16} + 1\right) - \frac{k}{16}\Big)_{k=0,1,2,3}$

Diferansiyel Denklemler için Sayısal Yöntemler

- Euler Yöntemi
- Yüksek dereceden Taylor Yöntemi